

TRIGONOMETRIC FUNCTIONS

3.1 Introduction

Trigonometry

The word **trigonometry** is derived from the Greek words *trigon* (triangle) and *metron* (measure). It deals with the study of relationships between angles and sides of triangles.

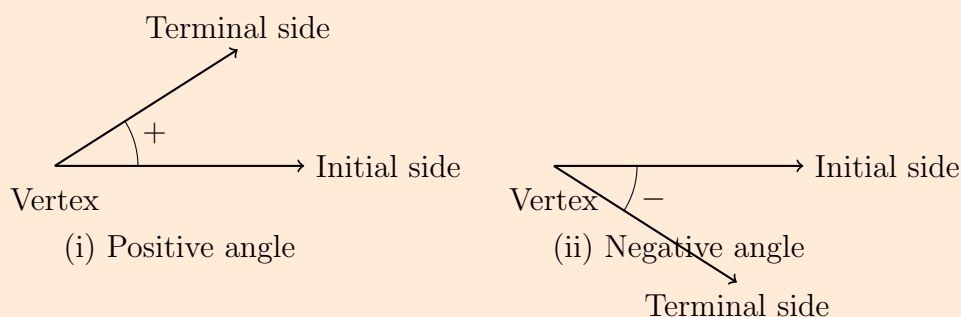
3.2 Angles

Angle

An angle is a measure of rotation of a ray about its initial point.

- Fixed point $\rightarrow$ **Vertex**
- Starting ray $\rightarrow$ **Initial side**
- Final ray $\rightarrow$ **Terminal side**

Positive and Negative Angles



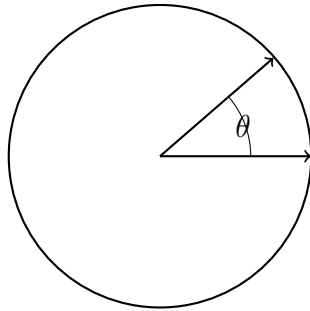
Anticlockwise rotation gives a positive angle, while clockwise rotation gives a negative angle.

3.2.1 Degree Measure

Degree Measure

If a rotation is $\frac{1}{360}$ of a complete revolution, the angle is said to have a measure of **one degree** (1°).

$$1^\circ = 60', \quad 1' = 60''$$



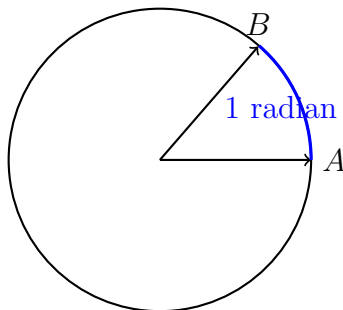
Angle in degree measure

3.2.2 Radian Measure

Radian Measure

An angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle is called **one radian**.

$$\theta = \frac{l}{r}, \quad l = r\theta$$



Arc length = radius

One complete revolution corresponds to 2π radians.

3.2.3 Relation Between Degree and Radian

Important Relation

$$180^\circ = \pi \text{ radians}$$

$$1^\circ = \frac{\pi}{180} \text{ radian}, \quad 1 \text{ radian} = \frac{180^\circ}{\pi}$$

Common Angle Values

Degree	30°	45°	60°	90°	180°
Radian	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π

3.3 Trigonometric Functions

In earlier classes, we studied trigonometric ratios of acute angles as the ratio of the sides of a right-angled triangle. We now extend these definitions to **angles of any magnitude** using the concept of the **unit circle**.

Unit Circle

Unit Circle

A circle with centre at the origin and radius equal to one unit is called the **unit circle**.

Let $P(a, b)$ be any point on the unit circle and $\angle AOP = x$ radians, measured anti-clockwise from the positive x -axis.

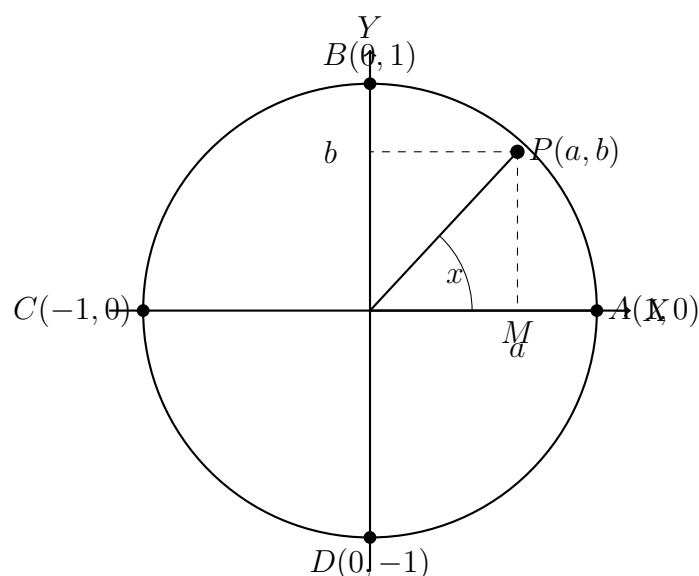


Fig. 3.6 : Unit Circle

Definition of Sine and Cosine

Definitions

If $P(a, b)$ is a point on the unit circle corresponding to angle x , then

$$\cos x = a, \quad \sin x = b$$

Since $\triangle OMP$ is a right-angled triangle,

$$OM^2 + MP^2 = OP^2 \Rightarrow a^2 + b^2 = 1$$

Hence, for every point on the unit circle,

$$\sin^2 x + \cos^2 x = 1$$

Quadrantal Angles

Quadrantal Angles

Angles which are integral multiples of $\frac{\pi}{2}$ are called **quadrantal angles**.

$$0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

For these angles:

Angle	$\cos x$	$\sin x$
0	1	0
$\frac{\pi}{2}$	0	1
π	-1	0
$\frac{3\pi}{2}$	0	-1
2π	1	0

Periodicity of Sine and Cosine

If we take one complete revolution from a point on the unit circle, we return to the same point.

Periodic Nature

$$\sin(x + 2\pi) = \sin x, \quad \cos(x + 2\pi) = \cos x$$

Also,

$$\begin{aligned} \sin x = 0 &\Rightarrow x = n\pi \\ \cos x = 0 &\Rightarrow x = (2n + 1)\frac{\pi}{2}, \quad n \in \mathbb{Z} \end{aligned}$$

Other Trigonometric Functions

Definitions of Other Functions

$$\begin{aligned}\csc x &= \frac{1}{\sin x}, & x &\neq n\pi \\ \sec x &= \frac{1}{\cos x}, & x &\neq (2n+1)\frac{\pi}{2} \\ \tan x &= \frac{\sin x}{\cos x}, & x &\neq (2n+1)\frac{\pi}{2} \\ \cot x &= \frac{\cos x}{\sin x}, & x &\neq n\pi\end{aligned}$$

Important Identities

$$\sin^2 x + \cos^2 x = 1$$

It follows that:

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Values of Trigonometric Functions

Values of Trigonometric Functions

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π
$\sin x$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	0
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	1
$\tan x$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	ND	0	0

ND = Not Defined

3.3.1 Sign of Trigonometric Functions

Let $P(a, b)$ be a point on the unit circle corresponding to angle x .

Even and Odd Properties

$$\cos(-x) = \cos x, \quad \sin(-x) = -\sin x$$

Since $-1 \leq a \leq 1$ and $-1 \leq b \leq 1$, we have

$$-1 \leq \cos x \leq 1, \quad -1 \leq \sin x \leq 1.$$

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Signs in Different Quadrants

Quadrant-wise Signs

Function	I	II	III	IV
$\sin x$	+	+	—	—
$\cos x$	+	—	—	+
$\tan x$	+	—	+	—
$\csc x$	+	+	—	—
$\sec x$	+	—	—	+
$\cot x$	+	—	+	—

3.3.2 Domain and Range of Trigonometric Functions

From the definition of sine and cosine functions, we observe that they are defined for all real numbers.

Basic Result

$$-1 \leq \sin x \leq 1, \quad -1 \leq \cos x \leq 1$$

Domain and Range:

Domain and Range

Function	Domain	Range
$\sin x$	$\mathbb{R}$	$[-1, 1]$
$\cos x$	$\mathbb{R}$	$[-1, 1]$
$\tan x$	$x \neq (2n+1)\frac{\pi}{2}$	$\mathbb{R}$
$\cot x$	$x \neq n\pi$	$\mathbb{R}$
$\sec x$	$x \neq (2n+1)\frac{\pi}{2}$	$(-\infty, -1] \cup [1, \infty)$
$\csc x$	$x \neq n\pi$	$(-\infty, -1] \cup [1, \infty)$

where $n \in \mathbb{Z}$.

Behaviour of Trigonometric Functions:

Behaviour in Quadrants

The behaviour of trigonometric functions depends on the quadrant in which the angle lies.

Function	I Quadrant	II Quadrant	III Quadrant	IV Quadrant
$\sin x$	increases $0 \rightarrow 1$	decreases $1 \rightarrow 0$	decreases $0 \rightarrow -1$	increases $-1 \rightarrow 0$
$\cos x$	decreases $1 \rightarrow 0$	decreases $0 \rightarrow -1$	increases $-1 \rightarrow 0$	increases $0 \rightarrow 1$
$\tan x$	$0 \rightarrow \infty$	$-\infty \rightarrow 0$	$0 \rightarrow \infty$	$-\infty \rightarrow 0$
$\cot x$	$\infty \rightarrow 0$	$0 \rightarrow -\infty$	$-\infty \rightarrow 0$	$0 \rightarrow \infty$
$\sec x$	$1 \rightarrow \infty$	$-\infty \rightarrow -1$	$-\infty \rightarrow -1$	$1 \rightarrow \infty$
$\csc x$	$1 \rightarrow \infty$	$1 \rightarrow \infty$	$-\infty \rightarrow -1$	$-\infty \rightarrow -1$

Remark

The symbols ∞ and $-\infty$ indicate unbounded behaviour and not actual values.

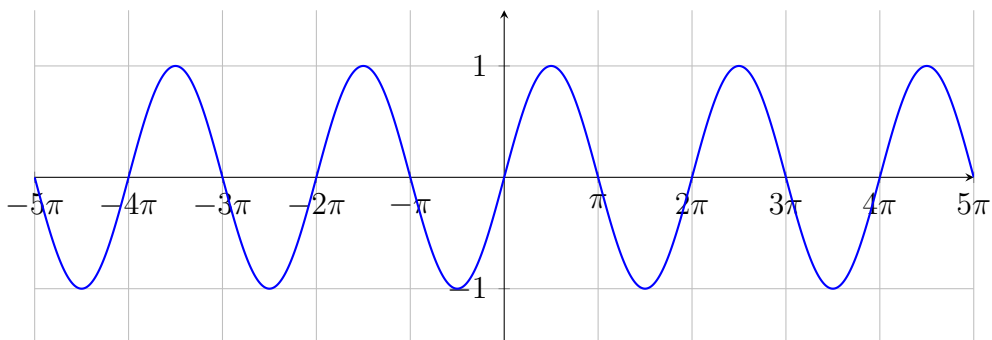
Periodicity:

Periodicity

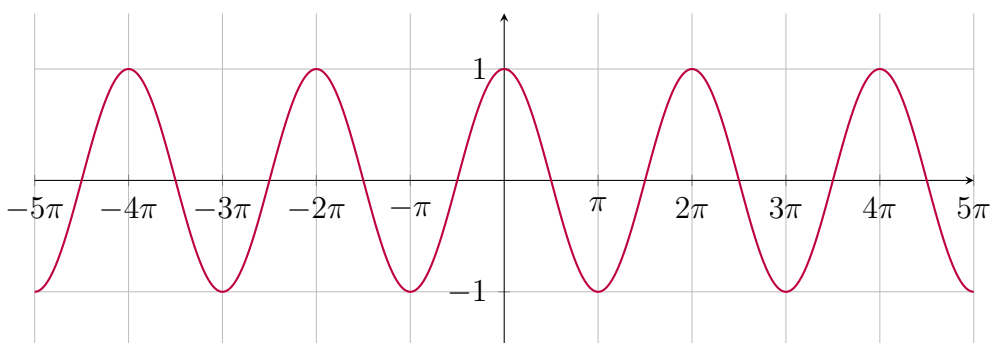
$$\sin(x + 2\pi) = \sin x, \quad \cos(x + 2\pi) = \cos x$$

$$\tan(x + \pi) = \tan x, \quad \cot(x + \pi) = \cot x$$

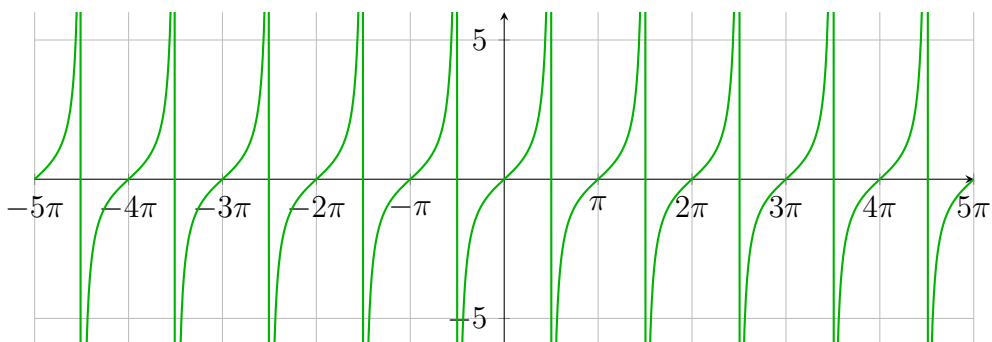
$$y = \sin x$$



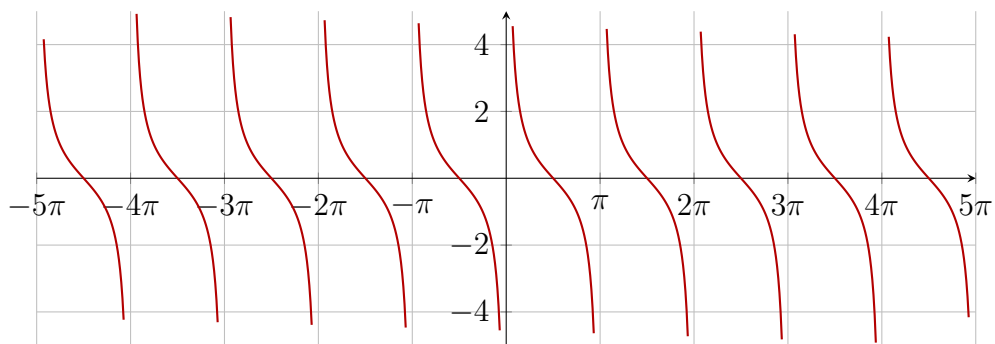
$$y = \cos x$$



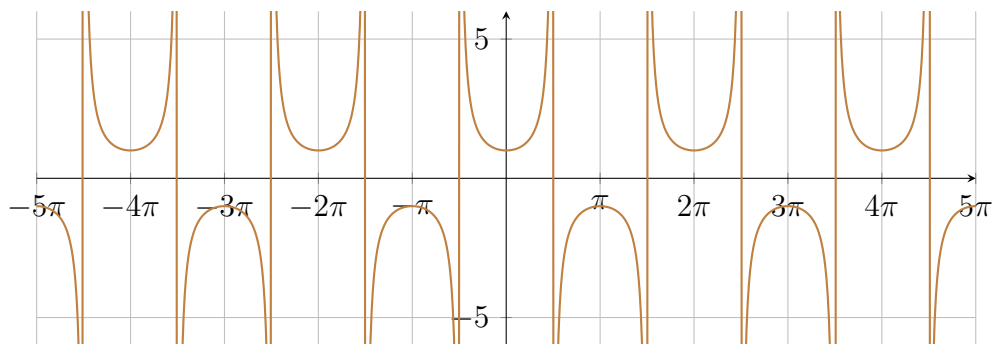
$$y = \tan x$$



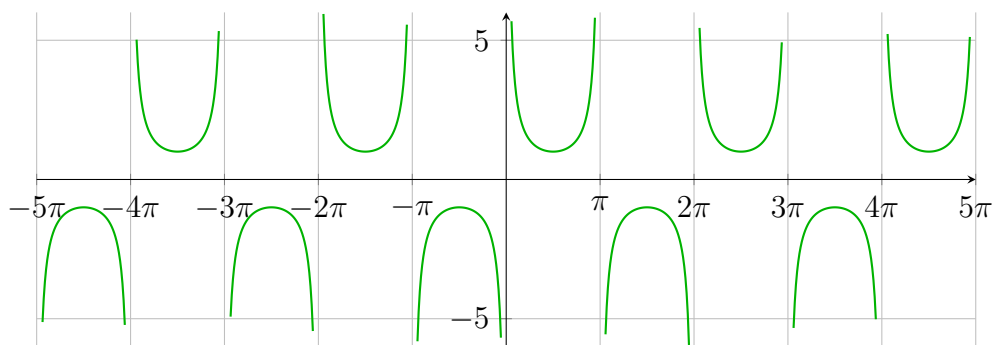
$$y = \cot x$$



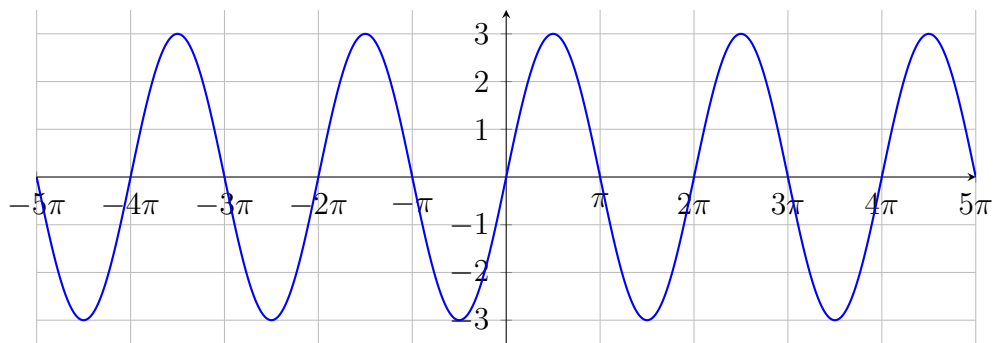
$$y = \sec x$$



$$y = \csc x$$



$$y = 3 \sin x$$



3.4 Trigonometric Functions of Sum and Difference of Two Angles

In this section, we derive expressions for trigonometric functions of the sum and difference of two angles. These results are called **trigonometric identities**.

Basic Identities

$$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x$$

Cosine Identities

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

Sine Identities

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

Allied Angle Identities

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x, \quad \sin\left(\frac{\pi}{2} - x\right) = \cos x$$

$$\cos(\pi - x) = -\cos x, \quad \sin(\pi - x) = \sin x$$

$$\cos(\pi + x) = -\cos x, \quad \sin(\pi + x) = -\sin x$$

$$\cos(2\pi - x) = \cos x, \quad \sin(2\pi - x) = -\sin x$$

Tangent and Cotangent Formulae

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\cot(x + y) = \frac{\cot x \cot y - 1}{\cot x + \cot y}$$

$$\cot(x - y) = \frac{\cot x \cot y + 1}{\cot y - \cot x}$$

Double and Triple Angle Formulae

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

$$\sin 2x = 2 \sin x \cos x = \frac{2 \tan x}{1 + \tan^2 x}$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

3.4 Trigonometric Functions of Sum and Difference of Two Angles

Basic Results

1. $\sin(-x) = -\sin x$
2. $\cos(-x) = \cos x$

Sum and Difference Formulae

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

Tangent Formulae

If none of the angles is an odd multiple of $\frac{\pi}{2}$, then

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}, \quad \tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}.$$

Multiple Angle Formulae

Double Angle Formulae

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

Triple Angle Formulae

$$\sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

$$\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$$

Sum-to-Product Formulae

Sum and Difference Conversion

$$\cos x + \cos y = 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2}$$

$$\cos x - \cos y = -2 \sin \frac{x+y}{2} \sin \frac{x-y}{2}$$

$$\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}$$

$$\sin x - \sin y = 2 \cos \frac{x+y}{2} \sin \frac{x-y}{2}$$

Examples

Example 1

Prove that:

$$3 \sin \frac{\pi}{6} \sec \frac{\pi}{3} - 4 \sin \frac{5\pi}{6} \cot \frac{\pi}{4} = 1$$

Solution:

$$= 3 \cdot \frac{1}{2} \cdot 2 - 4 \cdot \frac{1}{2} \cdot 1 = 3 - 2 = 1$$

Example 2

Find the value of $\sin 15^\circ$.

Solution:

$$\begin{aligned}\sin(45^\circ - 30^\circ) &= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3} - 1}{2\sqrt{2}}\end{aligned}$$

Example 3

Find the value of $\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ$.

Solution:

$$= \sin(30^\circ + 60^\circ) = \sin 90^\circ = 1$$

Example 4

Prove that $\sin x(1 + \tan^2 x) = \sec x \tan x$.

Solution:

$$\sin x(1 + \tan^2 x) = \sin x \sec^2 x = \frac{\sin x}{\cos^2 x} = \sec x \tan x$$

Example 5

Find $\cos(-210^\circ)$.

Solution:

$$\cos(-210^\circ) = \cos 210^\circ = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$$

Example 6

If $\sin x = \frac{3}{5}$ and x lies in the first quadrant, find $\cos x$.

Solution:

$$\cos x = \sqrt{1 - \sin^2 x} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$$

Example 7

Prove that $\frac{1 - \tan^2 x}{1 + \tan^2 x} = \cos 2x$.

Solution:

$$\frac{1 - \tan^2 x}{1 + \tan^2 x} = \frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin^2 x} = \cos 2x$$

Example 8

Find the value of $\sin 75^\circ$.

Solution:

$$\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Example 9

Evaluate $\tan 225^\circ$.

Solution:

$$\tan(180^\circ + 45^\circ) = \tan 45^\circ = 1$$

Example 10

Prove that $\sin 3x = 3 \sin x - 4 \sin^3 x$.

Solution:

$$\begin{aligned}\sin(2x + x) &= \sin 2x \cos x + \cos 2x \sin x \\ &= 2 \sin x \cos^2 x + (1 - 2 \sin^2 x) \sin x = 3 \sin x - 4 \sin^3 x\end{aligned}$$

Example 11

Find the value of $\cos\left(\frac{\pi}{2} - x\right)$.

Solution:

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x$$

Example 12

If $\tan x = \frac{5}{12}$, find $\sin x$ and $\cos x$.

Solution:

$$\sin x = \frac{5}{13}, \quad \cos x = \frac{12}{13}$$

Example 13

Prove that $\sin^2 x + \cos^2 x = 1$.

Solution: Divide numerator and denominator of $\tan^2 x + 1 = \sec^2 x$ by $\sec^2 x$.

$$\sin^2 x + \cos^2 x = 1$$

Example 14

Evaluate $\tan\left(-\frac{\pi}{4}\right)$.

Solution:

$$\tan(-x) = -\tan x \Rightarrow \tan\left(-\frac{\pi}{4}\right) = -1$$

Example 15

Find $\cos 3x$ in terms of $\cos x$.

Solution:

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

Example 16

Prove that $\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$.

Solution:

$$\tan(x + y) = \frac{\sin(x + y)}{\cos(x + y)} = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

Example 17

Evaluate $\sin 660^\circ$.

Solution:

$$\sin 660^\circ = \sin(660^\circ - 360^\circ) = \sin 300^\circ = -\frac{\sqrt{3}}{2}$$

Exercise — Trigonometry

Exercise A

Q.1. Evaluate $\sin 75^\circ$.

Q.2. Find the value of $\cos 210^\circ$.

Q.3. If $\sin x = \frac{5}{13}$ and x lies in the first quadrant, find $\cos x$.

Q.4. Prove that:

$$(1 + \tan^2 x)(1 - \sin^2 x) = 1$$

Q.5. Evaluate $\tan\left(-\frac{\pi}{3}\right)$.

Q.6. Find the value of $\sin\left(\frac{\pi}{2} - 30^\circ\right)$.

Q.7. If $\tan x = \frac{3}{4}$ and x lies in the third quadrant, find $\sin x$.

Q.8. Prove that:

$$\frac{1 - \cos 2x}{1 + \cos 2x} = \tan^2 x$$

Q.9. Find the value of $\cos\left(\frac{7\pi}{6}\right)$.

Q.10. Evaluate $\sin 660^\circ$.

Q.11. Prove that:

$$\sin(x + y) + \sin(x - y) = 2 \sin x \cos y$$

Q.12. Find $\cos 3x$ in terms of $\cos x$.

Q.13. If $\sec x = \frac{13}{5}$ and x lies in the fourth quadrant, find $\tan x$.

Q.14. Evaluate $\tan 225^\circ$.

Q.15. Prove that:

$$\cos x + \cos y = 2 \cos \frac{x + y}{2} \cos \frac{x - y}{2}$$

Answers:

Answers

Ans.1. $\frac{\sqrt{6} + \sqrt{2}}{4}$

Ans.2. $-\frac{\sqrt{3}}{2}$

Ans.3. $\frac{12}{13}$

Ans.4. L.H.S. = R.H.S.

Ans.5. $-\sqrt{3}$

Ans.6. $\frac{1}{2}$

Ans.7. $-\frac{3}{5}$

Ans.8. L.H.S. = R.H.S.

Ans.9. $-\frac{\sqrt{3}}{2}$

Ans.10. $-\frac{\sqrt{3}}{2}$

Ans.11. $2 \sin x \cos y$

Ans.12. $4 \cos^3 x - 3 \cos x$

Ans.13. $-\frac{12}{5}$

Ans.14. 1

Ans.15. $2 \cos \frac{x+y}{2} \cos \frac{x-y}{2}$